

Change Log — Data Cleaning & Preparation

DecodeLabs Internship | Project 1 | dataset.csv

Stakeholders need to know WHAT was changed and WHY. This log documents every modification made to the raw dataset during the cleaning process.

Change Records

Change ID	Description	Impact	Status
CR001	Imputed missing CouponCode values with 'No Coupon' category (median/mode not used, since missing means no coupon was applied, not a numeric gap)	Preserved 309 records (25.75% of rows) without deletion	Resolved
CR002	Checked for duplicate OrderIDs and full-row duplicates; none found	Verified 0% duplicate identifiers across 1,200 records	Verified
CR003	Standardized Date column to ISO 8601 format (YYYY-MM-DD)	Verified 0% invalid/unparseable dates across 1,200 records	Resolved
CR004	Trimmed whitespace and applied consistent title casing to Product, PaymentMethod, OrderStatus, ReferralSource, ShippingAddress	Eliminated case/whitespace inconsistency risk across 5 text columns	Resolved
CR005	Trimmed CouponCode (case preserved, since it is a code, not display text)	Preserved code integrity (e.g. SAVE10 stayed SAVE10)	Resolved
CR006	Rounded UnitPrice and TotalPrice to 2 decimal places	Enforced consistent numeric precision across 1,200 records	Resolved

Verification Gate

Requirement	Result	Status
Duplicate unique identifiers (OrderID)	0 out of 1,200 (0.00%)	PASS
Incorrectly formatted dates	0 out of 1,200 (0.00%)	PASS
Remaining missing values	0 out of 1,200 rows	PASS

Tool used: Python (pandas) — chosen for scalability and reproducibility.

Dataset: 1,200 rows x 14 columns (OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, TotalPrice)